

UNIVERSIDADE FEDERAL DE VIÇOSA

**IMPACT OF NEWS ON THE BRAZILIAN FINANCIAL MARKET
USING A LARGE LANGUAGE MODEL (LLM)**

Arthur Bastos Machado de Oliveira

Student ID: 101598

ADVISOR: Prof. Luciano Dias de Carvalho

VIÇOSA – MG

JULY 2026

Arthur Bastos Machado de Oliveira

**IMPACT OF NEWS ON THE BRAZILIAN FINANCIAL MARKET
USING A LARGE LANGUAGE MODEL (LLM)**

Monograph presented to the Department of
Economics of the Universidade Federal de Viçosa
as part of the requirements for the degree of
Bachelor in Economic Sciences.

APPROVED: ____ of _____ of ____.

Committee Member

Committee Member

Luciano Dias de Carvalho (Advisor)

VIÇOSA – MG

JUNE 2026

The opinions expressed in this work are the sole responsibility of the author.

ABSTRACT

OLIVEIRA, Arthur Bastos Machado de, B.Sc., Universidade Federal de Viçosa, June 2026. Impact of News on the Brazilian Financial Market Using a Large Language Model (LLM). Advisor: Luciano Dias de Carvalho.

This work investigates the effectiveness of open-source Large Language Models (LLMs) in classifying the sentiment of Portuguese-language financial news and its association with the short-term returns of stocks listed on the B3 exchange. A total of 10,108 headlines about ten of the leading companies listed on the Brazilian exchange, published between 2023 and 2024, were collected and classified using the Llama 3.1 8B Instruct model under a zero-shot inference regime. The association between the extracted sentiment and price behavior was assessed across four complementary layers of evidence: an event study based on the Market Model, which found a mean Cumulative Abnormal Return of +0.19% for positive news and -0.61% for negative news (both $p < 0.001$); a directional accuracy analysis, which obtained 54.74% accuracy against a baseline of 51.64% (McNemar, $p = 0.0021$); a logistic regression with firm fixed effects, in which sentiment remained significant (odds ratio of 1.20; $p < 0.001$) even after clustering standard errors by date; and a long/short portfolio backtest, which produced a Sharpe ratio of 3.80, higher than market benchmarks, though qualified by methodological caveats regarding the absence of transaction costs. The results indicate that the sentiment extracted by an intermediate-scale LLM carries statistically detectable informational content about subsequent returns, of economically modest magnitude, contributing to the scarce empirical literature on the use of LLMs for financial news in Portuguese.

Keywords: Large Language Models; Sentiment Analysis; Brazilian Financial Market; Event Study.

1. INTRODUCTION

The dynamics of financial markets are, in essence, a continuous process of information assimilation. As postulated by the Efficient Market Hypothesis (EMH) in its semi-strong form (FAMA, 1970), asset prices should instantly reflect all publicly available information. Historically, this price-adjustment mechanism depended on the human capacity to interpret news, earnings reports, monetary-policy decisions, and corporate announcements. However, the advent of the Big Data era radically transformed this landscape, generating a volume of unstructured data (mostly text) that far exceeds the human capacity for timely manual processing.

In this context, the finance literature has increasingly turned to quantitative methods capable of extracting trading signals from text. The seminal work of Tetlock (2007) empirically demonstrated that media pessimism negatively influenced stock prices and increased market volatility. Initially, these analyses relied on rudimentary Natural Language Processing (NLP) techniques, such as the Bag-of-Words approach and the use of polarity dictionaries, with the lexicon developed by Loughran and McDonald (2011) as the industry gold standard. These authors identified that words with negative connotations in common language (such as “cost” or “liability”) do not always carry the same sentiment in financial jargon, requiring adapted word lists.

Despite representing an advance, dictionary- and word-count-based approaches have severe limitations. They fail to capture context, irony, complex negation, and the semantic nuances that make up the interpretation of a text. A sentence such as “The company managed to avoid the expected drop in profits” would be classified as negative by traditional algorithms because of the words “drop” and “avoid,” when in reality it carries a positive sentiment of exceeding expectations.

The recent revolution in Artificial Intelligence, driven by the Transformer architecture introduced by Vaswani et al. (2017), enabled the emergence of Large Language Models (LLMs). Unlike their predecessors, models such as BERT (Bidirectional Encoder Representations from Transformers) and, more recently, the GPT (OpenAI) and Llama (Meta) families, possess “attention” mechanisms that allow them to understand the semantic relationship between words, interpreting text with a competence close to that of humans. Recent studies, such as Lopez-Lira and Tang (2023), suggest that advanced LLMs

significantly outperform traditional methods in predicting stock returns based on news headlines, indicating a paradigm shift in automated financial analysis.

However, a relevant gap is observed in the empirical literature. The vast majority of studies concentrate on the U.S. market and the English language. Applying these technologies in emerging markets, such as Brazil, faces additional challenges, such as the scarcity of labeled financial datasets in Portuguese and the specificities of local political and economic volatility. Moreover, many studies rely on proprietary, high-cost models (such as GPT-4), so there is a need to validate the effectiveness of open-source models, which offer greater transparency and scientific reproducibility.

The present work seeks to fill this gap by investigating the following research question: how effective are open-source Large Language Models (LLMs) at classifying the sentiment of Portuguese-language financial news, and what is their impact on the short-term returns of stocks listed on the B3? To answer this question, this study incorporates different statistical methods: first, the statistical significance of the Cumulative Abnormal Return (CAR) is estimated via an Event Study; then an accuracy analysis with a McNemar test and a logistic regression with firm fixed effects; and finally a backtest with the construction of a long/short portfolio and calculation of the Sharpe ratio. A total of 10,798 headlines related to ten of the leading companies listed on the B3 between 2023 and 2024 were collected and processed. Sentiment classification was performed using the Llama 3.1 8B Instruct model, instructed to act as a financial analyst, and the results were compared against the abnormal returns of the respective stocks to verify the existence of a statistical association between the extracted sentiment and price behavior.

The general objective, therefore, is to investigate the effectiveness of an open-source Large Language Model (LLM) in classifying the sentiment of Portuguese-language financial news and to assess its association with the short-term returns of stocks listed on the B3, through an empirical design across multiple layers of evidence.

The work is structured as follows: after this introduction, Section 2 presents the theoretical framework on market efficiency, behavioral finance, and the evolution of NLP in finance. Section 3 details the methodology of data collection, the architecture of the language model used, and the econometric model. Section 4 presents the analysis of results and, finally, Section 5 offers the concluding remarks.

2. THEORETICAL FRAMEWORK

This section establishes the theoretical foundation and empirical evidence that support the central hypothesis of this work: that the sentiment extracted from financial news by Large Language Models (LLMs) is associated with stock returns in the capital market.

2.1. THE INFORMATION PARADOX: EMH VS. SENTIMENT

The starting point of market analysis is the Efficient Market Hypothesis (EMH), proposed by Fama (1970). Grounded in assumptions of Classical Economics, such as the strict rationality of agents, the EMH is traditionally divided into three forms, according to the information set presumed to be reflected in prices. In the weak form, prices reflect all information contained in historical price series, making technical analysis ineffective at generating consistent abnormal returns. In the semi-strong form — the one relevant to this work — prices incorporate, beyond the price history, all publicly available information, including financial statements, news, and corporate announcements, such that no agent could obtain systematic abnormal returns by processing already-public information. In the strong form, prices would reflect even insider information, a hypothesis whose violation is widely documented and is not the object of this study.

It is important to note, following Fama (1991) and the subsequent literature, that every empirical test of the EMH is necessarily a joint test: rejecting the null hypothesis that prices reflect all available information is inseparable from rejecting (or not) the asset-pricing model used to define what a “normal” return would be — in the case of this work, the Market Model (Section 3.4). A statistically significant abnormal return may, in principle, reflect either a genuine informational inefficiency or a misspecification of the expected-return model. This is one of the reasons why the event-study literature treats results as evidence of association between the event and the return, and not as definitive proof of market inefficiency.

Periods of high volatility throughout the 1980s revealed important limitations of the EMH, as demonstrated by Shiller (2003), from which the Theory of Behavioral Finance emerged (Thaler; Kahneman & Tversky), which argues that the behavior of economic agents is driven by cognitive and emotional biases, resulting in overreaction or underreaction to news events.

Complementarily, the literature on noise trader models (De Long et al., 1990) and limits to arbitrage (Shleifer; Vishny, 1997) offers a theoretical mechanism for pricing inefficiencies to persist even in the presence of rational agents: arbitraging a price deviation requires capital, time, and willingness to bear risk, so rational agents may prefer not to fully

correct a pricing error caused by behaviorally biased investors, especially over short horizons like the one studied here (a two-day window). In this scenario, investor sentiment, frequently captured by the tone of the media, becomes a fundamental variable for explaining price movements. The existence of consistent and statistically significant abnormal returns after the release of news (the post-earnings-announcement drift or post-event drift) is one of the main pieces of evidence of inefficiency, justifying the search for predictive factors beyond fundamentals. As will be discussed in Section 4, the asymmetric magnitude of the abnormal returns associated with positive and negative news found in this study is itself coherent with this literature on asymmetric reaction and with the loss-aversion mechanism proposed by Kahneman and Tversky (1979).

2.2. THE EVOLUTION OF TEXT ANALYSIS AND LLMS IN FINANCE

The attempt to quantify media sentiment evolved in three main stages:

2.2.1. Dictionaries and word counting (the pre-LLM era)

The seminal work of Loughran and McDonald (2011) transformed textual analysis in finance by creating a specific lexicon, adjusting common words to the financial context. Tetlock's (2007) study, which correlated media pessimism with price declines, used text analysis as a proxy for sentiment. These methods, although efficient, have the critical limitation of being blind to context. They cannot distinguish the positive impact of “avoiding a profit drop” from the negative impact of “profit drop,” relying solely on word counts.

2.2.2. Contextual models and LLMs

The contextual limitation was overcome by the emergence of models based on the Transformer architecture (VASWANI et al., 2017). LLMs such as BERT, GPT, and, more recently, Llama, use “attention” mechanisms to understand the dependency and meaning of words within a sentence. The global empirical literature demonstrates the predictive superiority of these architectures. The study by Kirtac and Germano (2024), comparing LLMs (such as OPT, BERT, and Llama 3) against the Loughran-McDonald dictionary model on 965,000 financial news articles, found that the OPT model significantly outperformed the others, reaching a return-direction prediction accuracy of 74.4%, and that a long-short portfolio strategy based on its scores achieved a Sharpe ratio of 3.05, compared to 1.23 for the dictionary-based strategy.

In a similar direction, Lopez-Lira and Tang (2023) documented that GPT-4, when classifying news headlines as positive, negative, or neutral for the corresponding company, reaches hit rates of approximately 90% in identifying the direction of the initial market reaction, and that financial-evaluation capacity increases with model size — simpler models, such as Llama 2-7b, exhibit limited predictive capacity compared to larger-scale models. This evidence is particularly relevant to the present study, which employs an intermediate-scale model (Llama 3.1 8B), and will be revisited in the discussion of results in Section 4.

This evidence corroborates the effectiveness of using LLMs for the specific task of sentiment classification associated with returns, while at the same time suggesting that the magnitude of that effectiveness is sensitive to the scale of the model employed.

2.2.3. LLMs in complex documents

Beyond short news, LLMs have proven effective at summarizing lengthy documents. The work of Gupta (2023) demonstrates the value of LLMs in analyzing corporate Annual Reports, which reach more than 100 pages, using LLM-generated features in a Machine Learning model for quant trading and obtaining performance superior to the S&P 500 over the analyzed period. This evidence reinforces the potential of the LLM as a tool for extracting value from financial texts of different natures and lengths.

2.3. EMPIRICAL EVIDENCE IN THE BRAZILIAN MARKET (B3)

The application of textual analysis on the B3 is scarcer but equally relevant. Dias and Malaquias (2021), using the event-study methodology on a sample of 83 Brazilian companies and 236 news items between 2012 and 2019, found that the disclosure of news related to possible involvement in corruption generates a negative and statistically significant abnormal return, mainly on the disclosure date, showing that events of a not strictly financial nature (such as ethical scandals) have a relevant effect on the market value of companies listed on the B3. Padula and Albuquerque (2018) also found a negative and significant relationship between the disclosure of information related to government corruption and the valuation of state-owned enterprises in the Brazilian capital market.

These two works share with the present study the same underlying methodological structure — the event study with a single-factor Market Model — although they classify the event of interest in a binary and manual manner (presence or absence of corruption news), and not by means of a language model applied to a broad flow of news of diverse nature. The

magnitude of the negative abnormal returns they report serves, in Section 4, as a benchmark for assessing the plausibility of the effects estimated here.

The need to test Llama 3.1 in the Brazilian context is based on the premise that the effectiveness of language models may vary according to the language, the specific domain (financial news), and the market structure, justifying the focus on computing the Abnormal Return (AR) for B3 assets.

3. METHODOLOGY

The methodology chosen to analyze the impact of news on asset prices was the event study. This method aims to estimate the impact of an event on a variable by computing the difference between expected and observed returns after the occurrence of the phenomenon.

3.1. DATA AND SAMPLES

Data were collected on 10 of the largest companies on the Brazilian stock exchange. The construction of the sample was segmented into two main sources: textual data (news) and financial data (quotes).

The 2023–2024 window corresponds to a period of macroeconomic and political transition relevant to the interpretation of the results. It was a period marked by the return to “normality” of the markets after the shock of the Covid-19 pandemic and its relative stabilization throughout 2022. Early 2023 marked the change of federal government, with direct repercussions on state-owned enterprises or those strongly interconnected with economic policy — in particular Petrobras, whose fuel-pricing and dividend-distribution policies were the subject of reformulations and intense press coverage throughout the period, and Banco do Brasil, whose governance and credit strategy were also under public scrutiny. The period also coincides with a cycle of contractionary monetary policy followed by easing: the SELIC rate, which serves in this work as a proxy for the risk-free asset, remained at an elevated level throughout 2023 and began a downward trajectory of cuts starting in the second half of the year, which affects the cost of capital and the relative attractiveness of interest-rate-sensitive sectors such as banking (Bradesco, Banco do Brasil, Itaú, BTG Pactual, Itaúsa) and consumer goods (Ambev).

The selection of the ten companies obeyed four joint criteria. First, representativeness in the market index: all selected companies are among those with the greatest weight in the Ibovespa over the analyzed period, which ensures that the market proxy used in the Market

Model (the Ibovespa itself) has a direct and relevant relationship with the studied assets, avoiding the use of a benchmark poorly correlated with the securities of interest. Second, liquidity: the ten stocks present high average daily trading volume and constant presence on the trading floor, a necessary condition for adjusted prices to reflect timely market information and for the computation of daily returns not to be distorted by illiquidity or stale prices. Third, sectoral diversification: the sample covers oil and gas (Petrobras), the banking and financial sector (Bradesco, Banco do Brasil, Itaú, BTG Pactual, Itaúsa), consumer goods (Ambev), public utilities/electricity (Eletrobras), mining (Vale), and capital goods/industry (WEG), reducing the chance that the results are an artifact of a single sector of the economy. Fourth, minimum volume of press coverage: companies with little media presence would make it unfeasible to apply the four proposed testing layers, especially the logistic regression and the portfolio backtest, which depend on a minimum number of events per asset to yield statistical inference with adequate power. This criterion, at the same time, explains why smaller companies or those with less media coverage, even if relevant on the exchange, were not included in this first application of the methodology.

These criteria imply, as an acknowledged trade-off, the concentration of the sample in large, high-visibility companies (Section 4.2.2), but they ensure the statistical feasibility of the four-layer design and comparability with the market proxy used.

Eletrobras, in turn, was undergoing a process of corporate consolidation following its privatization, completed in 2022, which kept the company under constant market monitoring regarding changes in governance and capital structure. Vale remained under monitoring related to environmental liabilities and international iron-ore prices, while WEG, the only representative of the industrial capital-goods sector in the sample, mainly reflected operational and earnings news, with substantially lower coverage volume than the others.

From the standpoint of the external validity of the results, it is relevant to note that the period does not include large-magnitude systemic shocks, such as those observed during the Covid-19 pandemic, which favors the interpretation that the effects estimated here reflect the “normal” dynamics of pricing corporate news and not the market's response to an extreme macroeconomic event. On the other hand, the concentration of press coverage around Petrobras, Bradesco, and Banco do Brasil (Section 3.1.1) is partially associated with this context of greater political and regulatory exposure of these three companies during the period, which is revisited as a limitation in Section 4.2.2.

3.1.1. Collection of textual data (news)

The news sample was collected via RSS Feeds indexed by Google News, using the pygooglenews library (Python). The collection period spans 24 months, from January 1, 2023, to December 31, 2024.

To maximize relevance and avoid noise (e.g., headlines with the word “Vale” unrelated to the mining company), a boolean (OR) search strategy was used, focusing the search on terms appearing in the title (intitle:). A total of 10,798 relevant news items were found, concentrated in the companies Petrobras, Bradesco, and Banco do Brasil, which make up almost 62% of the entire sample, as described in Table 1. The main news sources are described in Table 2.

Table 1: Keywords and number of news items per company

Ticker	Keywords	Total Volume	% of Total
PETR4	Petrobras, PETR4, PETR3	2,398	22.21
BBDC4	Bradesco, BBDC4, Banco Bradesco	2,260	20.93
BBAS3	Banco do Brasil, BBAS3	2,034	18.84
ABEV3	Ambev, ABEV3	964	8.93
ELET3	Eletrobras, ELET3, ELET6	934	8.65
BPAC11	BTG Pactual, BPAC11, Banco BTG	636	5.89
ITUB4	Itaú Unibanco, ITUB4, Banco Itaú	627	5.81
VALE3	Vale S.A., VALE3, Vale mineradora	620	5.74
WEGE3	WEGE3, WEG Motores, Empresa WEG	188	1.74
ITSA4	Itaúsa, ITSA4	137	1.27
Total	—	10,798	100.00

Source: Research data.

Table 2: Top 10 News Sources

Source	Count
InfoMoney	691
Estadão E-Investidor	686
Money Times	332
suno.com.br	326
CNN Brasil	295
O Globo	261
Agência Brasil	252
Exame	234
Valor Econômico	220
Seu Dinheiro	209
Total (top 10)	3,506

Source: Research data.

Table 3 presents real examples of headlines effectively classified by the pipeline described in Section 3.2, selected for proximity to the mean CAR of their respective sentiment group (Table 5, Section 4.2), so as to illustrate the typical case of the sample and not its extreme-magnitude cases.

Table 3: Examples of headlines classified by the LLM and associated cumulative abnormal return

Company	Headline (original title, translated)	Sentiment (LLM)	CAR[T0, T+1]
BBAS3	Banco do Brasil posts profit of R\$8.6bn in Q1 — Seu Crédito Digital	POSITIVE	+0.19%
ITUB4	Itaú Unibanco quarterly profit rises 14.6% to R\$8.43bn — Safra	POSITIVE	+0.23%
PETR4	Petrobras (PETR4) earns R\$26.6bn in Q3, down 42.2% year-on-year and below consensus — InfoMoney	NEGATIVE	−0.58%
BBDC4	Bradesco Pix down: users report app instability — TechTudo	NEGATIVE	−0.61%
BBAS3	Banco do Brasil to sponsor the SLS Championship Tour — Marcas pelo Mundo	NEUTRAL	−0.02%
ITUB4	Itaú (ITUB4) proposes incorporation of Hipercard and calls Extraordinary Meeting — suno.com.br	NEUTRAL	+0.00%

Source: Research data.

Table 3 illustrates, with concrete cases, the pattern examined statistically in Section 4.2: clearly positive or negative headlines precede abnormal returns in the corresponding direction, while headlines of an informational or operational nature — such as corporate meeting calls or sponsorships — are predominantly classified as neutral and associated with CAR close to zero.

3.1.2. Collection of financial data (quotes)

The supporting financial data, needed to compute returns and the benchmark, were obtained from two sources:

Asset and index quotes: the daily historical series of adjusted close prices (Adj Close) of the 10 companies and of the Ibovespa market index (^BVSP) were extracted via the Yahoo Finance API (yfinance). This source is widely accepted in empirical research due to its reliability and its inclusion of automatic adjustments for corporate actions (dividends, interest on equity). It should be noted that Eletrobras changed its trading symbol from ELET3 to AXIA3 at the end of 2025; the historical prices for the 2023–2024 period used in this study were obtained under the current symbol, which Yahoo Finance keeps linked to the asset's full historical series, preserving correspondence with the news collected under the previous denomination (ELET3).

Risk-free asset: the proxy for the risk-free return in Brazil was the Daily SELIC Rate (CDI Rate), obtained from the API of the Time Series Management System (SGS) of the Central Bank of Brazil (code 12). This rate is fundamental to the computation of excess returns in the Market Model (Equation 1).

3.2. SENTIMENT ANALYSIS WITH AN LLM (LLAMA 3.1 8B)

The classification of the 10,798 headlines was performed using a Large Language Model (LLM), adopting the approach of inference and classification without supervised training (zero-shot learning).

3.2.1. Inference architecture and infrastructure

The selected model was Meta Llama 3.1 8B Instruct (8 billion parameters). This choice is justified by its performance in reasoning and instruction-following tasks (a characteristic of the Instruct version) and by being an open-weights (open-source) model, with lower computational cost than proprietary models such as GPT-4, directly addressing the research question posed in Section 1.

Inference was executed through the Groq API, a provider specialized in hosting open-weights language models on proprietary hardware (LPU – Language Processing Unit), optimized for high-throughput inference. The Groq API is compatible with the OpenAI API specification, which allowed the classification pipeline to be integrated using a standard client library. It is worth emphasizing that the use of a hosted inference API does not compromise the open-source nature of the experiment: the model weights remain the same ones publicly released by Meta, with Groq responsible only for the hosting and execution layer, analogous to the use of a cloud instance with a dedicated GPU.

3.2.2. Prompt engineering strategy and batch processing

Rather than training the model for the specific task (which would be fine-tuning), the Prompt Engineering strategy was chosen, defining the persona, the context, and the desired output format. This turns the LLM into a text classifier with financial-domain knowledge. Inference was configured to be deterministic (parameter temperature=0), which reduces randomness in the response and tends to make the same headline produce the same sentiment result across repeated runs.

Due to rate-limiting constraints imposed by the API provider, classification was implemented with batch processing: each request submitted a set of 15 headlines

simultaneously, with the model returning a vector of classifications in the same input order, in JSON format. Table 4 illustrates the logical structure of the prompt used for each individual headline within the batch.

Table 4: Prompt Structure for Sentiment Classification (Llama 3.1 8B)

Component	Content	Function
System Role	“You are an experienced financial analyst specialized in the Brazilian market (B3).”	Defines the persona and the knowledge domain.
User Content	Headline: “{title}” about the company {company}.	Injects the raw data and the trading context.
Constraint	“Respond only with a JSON array containing, in the same order, one of the labels: POSITIVE, NEGATIVE, or NEUTRAL, with no explanation.”	Forces the structured output required by the processing pipeline.

Source: Prepared by the author.

3.2.3. Encoding of the sentiment variable

The model's output variable is categorical, taking three possible values: POSITIVE, NEUTRAL, or NEGATIVE. For statistical analysis purposes, this variable was numerically encoded as: 1 if positive; 0 if neutral (headlines with no clear impact); −1 if negative. This encoding allows the Sentiment variable (S) to be used directly in the association tests with the Cumulative Abnormal Return (CAR), presented in Section 4.

3.2.4. Handling of classification failures

Of the total of 10,798 headlines submitted to the model, 690 (6.4% of the sample) could not be classified after successive reprocessing attempts, owing to formatting failures in the response returned by the API (for example, responses that did not constitute a valid JSON array or that contained a number of elements different from expected). These failures were found to be distributed proportionally across the ten companies and throughout the entire collection period, with no identifiable concentration in any specific ticker or date interval, which reduces the probability of systematic bias arising from the exclusion. These observations were removed from the sample without replacement, resulting in a final analytical sample of 10,108 classified headlines, used throughout the analysis presented in Section 4.

3.3. EVENT DEFINITION AND TIME WINDOW

The event (T0) is defined as the publication date of the headline, classified by the LLM. To ensure the pertinence of the data, events are mapped as follows: publications during trading hours (9:30 a.m. to 5:00 p.m.) have their impact measured on the same day and the

following business day; publications after the close or on weekends have their impact measured starting from the first subsequent business day. In practice, more than 97% of the headlines in the sample carry a publication time standardized by the RSS aggregator (between 4:00 and 5:00 a.m. Brasília time), prior to the market open — which simplifies the mapping rule, causing the overwhelming majority of events to follow the second criterion.

The analysis was divided into the following windows, following the procedures described by MacKinlay (1997) and detailed in Miller (2023): Estimation Window (L1): the period between 120 and 30 business days prior to the event, used to estimate the α and β parameters of the Market Model without contamination by the event itself; Event Window (L2): a two-day period — the event date (T0) and the following business day (T+1) — used to observe the immediate effect of the news.

3.4. COMPUTATION OF THE EXPECTED RETURN (MARKET MODEL)

To isolate the effect of the news from the general movement of the market, the return that would be expected of the stock in the absence of the event is estimated. The Market Model is used, a simplification of the CAPM that incorporates the risk-free asset.

3.4.1. Parameter estimation

The parameters of the linear relationship between the excess return of the asset and the excess return of the market are estimated by means of an OLS (Ordinary Least Squares) regression, using data from the Estimation Window:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{m,t} - R_{f,t}) + \varepsilon_{i,t} \quad (1)$$

Where: $R_{i,t}$ is the return of asset i on day t ; $R_{f,t}$ is the return of the risk-free rate (CDI) on day t ; $R_{m,t}$ is the return of the Ibovespa on day t (market proxy); $\varepsilon_{i,t}$ is the residual (error) of the model; α_i is the intercept — the return of the asset not explained by the market; β_i is the sensitivity coefficient (systematic risk) of asset i relative to the market.

3.5. COMPUTATION AND ACCUMULATION OF ABNORMAL RETURNS (AR AND CAR)

3.5.1. Expected return

After estimating the coefficients $\hat{\alpha}_i$ and $\hat{\beta}_i$ in the Estimation Window, the Expected Return for the days within the Event Window is computed as:

$$E[R_{i,t}] = \hat{\alpha}_i + \hat{\beta}_i (R_{m,t} - R_{f,t}) \quad (2)$$

3.5.2. Abnormal return (AR)

The Abnormal Return is the difference between the actually observed return and the expected return based on the historical relationship between the asset and the market:

$$AR_{i,t} = R_{i,t} - E[R_{i,t}] \quad (3)$$

3.5.3. Cumulative abnormal return (CAR)

To measure the total effect of the news over the Event Window, the Cumulative Abnormal Return is computed:

$$CAR_i(T0, T+1) = AR_{i,T0} + AR_{i,T+1} \quad (4)$$

Where T0 and T+1 define the Event Window used in this study, as established in Section 3.3.

3.5.4. Statistical test and significance

The central hypothesis test of this study compares the mean CAR between the groups of news classified as positive, neutral, and negative, formalized as:

Null Hypothesis (H0): the mean cumulative abnormal return of the group is equal to zero (mean CAR = 0).

Alternative Hypothesis (H1): the mean cumulative abnormal return of the group is different from zero (mean CAR $\neq$ 0).

Significance is tested by means of a one-sample t-test of the mean, verifying whether the mean CAR of each sentiment group is statistically different from zero, complemented by a one-sided t-test to verify whether the mean CAR of the positive-news group is statistically greater than the mean CAR of the negative-news group.

3.6. COMPLEMENTARY ANALYSIS: DIRECTIONAL ACCURACY AND LOGISTIC REGRESSION

Beyond the difference-in-means test presented in Section 3.5.4 — which assesses the association between sentiment and the magnitude of the abnormal return — a complementary analysis was conducted, aimed at the directional predictive capacity of the model, along the lines employed by Lopez-Lira and Tang (2023) and Kirtac and Germano (2024).

3.6.1. Binary reformulation and directional accuracy

For this analysis, observations classified as NEUTRAL were excluded, since a neutral classification does not, by definition, constitute a directional prediction — a procedure analogous to that adopted by Lopez-Lira and Tang (2023), who treat “Yes/No” classifications separately from “Unsure” ones in their methodology. The binary variable $Y=1$ was defined if $CAR[T0,T+1]>0$ and $Y=0$ otherwise, and the model's prediction $\hat{Y}=1$ if the news sentiment is POSITIVE and $\hat{Y}=0$ if NEGATIVE. Directional accuracy is then:

$$Accuracy = (number\ of\ correct\ predictions) / (total\ number\ of\ predictions) \quad (5)$$

For the accuracy obtained to have interpretive value, it is compared to a fair baseline, defined as the proportion of the majority class observed in the sample (and not 50% arbitrarily):

$$Baseline = \max(\% CAR>0, \% CAR\leq 0) \quad (6)$$

3.6.2. McNemar's test

To verify whether the difference between the model's accuracy and the baseline is statistically significant, and not the result of sampling variation, McNemar's test was applied (MCNEMAR, 1947), appropriate for comparing two classifiers evaluated on the same sample. The test considers only the cases in which the two classifiers diverge: let b be the number of cases in which only the model is correct, and c the number of cases in which only the baseline is correct. The test statistic is given by:

$$\chi^2 = (|b - c| - 1)^2 / (b + c) \quad (7)$$

which follows, under the null hypothesis of equivalence between the classifiers, a chi-square distribution with 1 degree of freedom.

3.6.3. Confusion matrix

Complementarily, the confusion matrix details the distribution of the model's hits and misses across the predicted categories (positive/negative) and observed categories (positive/non-positive CAR), allowing the identification of any systematic asymmetries in the direction of classification errors.

3.6.4. Logistic regression with firm fixed effects

Finally, a logistic regression model was estimated to investigate, in a more refined manner than raw accuracy, the relationship between sentiment and the probability of a

positive abnormal return, controlling for unobserved heterogeneity across firms by means of fixed effects:

$$P(CAR > 0) = 1 / (1 + e^{-(\beta_0 + \beta_1 \cdot S + \sum \gamma_j \cdot Firm_j)}) \quad (8)$$

Where S is the sentiment variable (encoded -1 or $+1$, per Section 3.2.3) and γ_j represents the fixed effect of firm j (with one reference category omitted for model identification). The coefficient β_1 is the parameter of interest, interpreted via the odds ratio e^{β_1} : the multiplicative factor on the odds of a positive abnormal return associated with positive news, holding the firm fixed.

The model fit is assessed by McFadden's Pseudo R^2 , a metric analogous to the R^2 of linear regression but suitable for models with a binary dependent variable.

For the covariance structure of the errors, clustering by publication date was adopted. Simultaneous clustering by date and by firm (double clustering, per Kirtac and Germano, 2024) would be the most general approach, since it addresses both the correlation between observations of different firms on the same day and the correlation between observations of the same firm on distinct dates. However, the asymptotic validity of clustered standard errors requires that the number of groups be sufficiently large — Cameron and Miller (2015) point to $G \geq 50$ as a minimum reference. With only 10 firms in the sample, clustering by firm would produce underestimated rather than conservative standard errors, compromising the robustness of the inference. Clustering by date, in turn, has approximately 500 distinct groups over the analyzed period, fully meeting the asymptotic requirement and correcting the main channel of dependence between observations: market shocks common to multiple firms on the same day. Functional-specification tests (link test) did not evidence misspecification of the adopted logistic form ($p=0.11$).

3.7. COMPLEMENTARY ANALYSIS: LONG/SHORT PORTFOLIO AND SHARPE RATIO

As an additional exercise of the economic — and not merely statistical — significance of the signal extracted by the LLM, the portfolio-backtest methodology employed by Kirtac and Germano (2024) was replicated on a reduced scale.

3.7.1. Portfolio construction

For each business day on which there was at least one non-neutral news item about any of the ten companies, a long position was built in the stocks whose aggregate sentiment

for the day was positive and a short position in the stocks whose aggregate sentiment was negative — with aggregate sentiment, when more than one news item for the same company occurs on the same day, defined by the mean of the variable S on that day. The daily return of the long/short, dollar-neutral portfolio was approximated by:

$$R_{portfolio} = (mean\ CAR_{long} - mean\ CAR_{short}) / 2 \quad (9)$$

It is worth highlighting a relevant methodological simplification: the return of each position is approximated by the $CAR[T_0, T+1]$ already computed in the event study (Section 3.5.3), attributing the entirety of the two-day return to the signal date — a usual procedure in academic event-based backtests, but one that does not correspond to a strict simulation of a portfolio with continuous daily rebalancing. Additionally, the exercise does not incorporate transaction costs, stock-borrow costs for the short positions, or liquidity restrictions — limitations revisited in the discussion of results in Section 4.4.

3.7.2. Sharpe ratio and standard error

The risk-adjusted performance of the strategy was evaluated by the annualized Sharpe ratio:

$$Sharpe = (mean\ R_{portfolio} / \sigma_{portfolio}) \times \sqrt{252} \quad (10)$$

Given that the estimate of the Sharpe ratio is subject to sampling error, its standard error was computed using the i.i.d. approximation of Lo (2002):

$$SE(Sharpe) \approx \sqrt{(1 + 0.5 \cdot Sharpe^2) / T} \quad (11)$$

where T is the number of days with an open position. This standard-error estimate presupposes independent and identically distributed returns, an assumption that may underestimate the real uncertainty if there is autocorrelation arising from the overlap between event windows of nearby dates — a caveat incorporated into the interpretation of results in Section 4.4.

The strategy's performance was compared to two benchmarks: the buy-and-hold portfolio of the Ibovespa and an equal-weighted basket of the ten sample stocks, both computed over the same period and with the same Sharpe-ratio methodology.

4. ANALYSIS OF RESULTS

4.1. DESCRIPTIVE STATISTICS

Of the 10,108 headlines that make up the final analytical sample (per the treatment described in Section 3.2.4), the classification produced by Llama 3.1 8B was distributed across the three sentiment categories as presented in Table 5.

Table 5: Distribution of news by sentiment classified by the LLM

Sentiment	N	% of Total
NEUTRAL	4,889	48.37
POSITIVE	2,738	27.09
NEGATIVE	2,481	24.55
Total	10,108	100.00

Source: Research data.

A predominance of news classified as neutral is observed (48.4%), followed by positive (27.1%) and negative (24.5%). This distribution is consistent with the nature of the news flow collected via Google News, which includes a large volume of institutional announcements, operational disclosures, and factual mentions with no clear sentiment content — and not only market events with a strong evaluative charge.

For all 10,108 observations it was possible to compute the Cumulative Abnormal Return (CAR), since the entirety of the events had at least 30 business days of data in the Estimation Window, as required by the methodology described in Section 3.4.

Figure 1 complements Table 6 (Section 4.2) by presenting the complete distribution of $CAR[T_0, T+1]$ in each sentiment group, and not only their means. The median CAR is close to zero in all three groups, but the box of the NEGATIVE group is shifted downward relative to that of the POSITIVE group, and its dispersion is the largest among the three — consistent with the largest standard deviation reported in Table 6 (2.97% in the NEGATIVE group, versus 2.38% in POSITIVE and 2.20% in NEUTRAL) and with the asymmetry discussed in Section 4.2.

Figure 1: Distribution of $CAR[T_0, T+1]$ by sentiment group

[Figure available in the repository: figures/car_boxplot.png] Source: Research data.

4.2. EVENT STUDY RESULTS: CUMULATIVE ABNORMAL RETURN (CAR)

Table 6 presents the results of the hypothesis test formulated in Section 3.5.4, comparing the mean $CAR[T_0, T+1]$ across the three sentiment groups.

Table 6: Hypothesis test results — $CAR[T_0, T+1]$ by sentiment group

Sentiment	N	Mean CAR (%)	Std. dev. (%)	t-stat	p-value	95% CI
POSITIVE	2,738	+0.1854	2.3766	4.0827	<0.001 ***	[+0.0964; +0.2745]

Sentiment	N	Mean CAR (%)	Std. dev. (%)	t-stat	p-value	95% CI
NEUTRAL	4,889	-0.0162	2.2020	-0.5144	0.607	[-0.0779; +0.0455]
NEGATIVE	2,481	-0.6078	2.9697	-10.1950	<0.001 ***	[-0.7247; -0.4909]

Note: *** $p < 0.01$. One-sample t-test of the mean (H_0 : mean CAR = 0). Source: Research data.

As a robustness analysis, an OLS regression of CAR on the sentiment dummies was estimated with standard errors clustered by date, controlling for potential cross-sectional dependence among the abnormal returns of simultaneous events. The coefficients remained statistically significant (NEGATIVE: $z = -3.17$, $p = 0.002$; POSITIVE vs. NEGATIVE: $z = 4.35$, $p < 0.001$), confirming that the conclusions of Section 4.2 are robust to the presence of intra-group correlation within each event date.

The results confirm, in a statistically robust manner, the central hypothesis of this work. The mean CAR of the POSITIVE news group (+0.19%) and of the NEGATIVE group (-0.61%) are both significantly different from zero at the 1% level, whereas the NEUTRAL group (-0.02%) shows no statistically significant difference from zero ($p = 0.607$) — behavior expected of a properly calibrated control group, which reinforces the validity of the classification produced by the LLM.

The one-sided t-test directly comparing the two extreme groups confirms that the mean CAR of positive news is statistically greater than that of negative news ($t = 10.6988$; $p < 0.001$).

A finding worth highlighting is the asymmetry in magnitude between the two effects: the mean CAR of the NEGATIVE group (-0.61%) is approximately 3.3 times larger, in absolute value, than the mean CAR of the POSITIVE group (+0.19%). This asymmetry is coherent with the Behavioral Finance literature discussed in Section 2.1, according to which investors tend to react more intensely to potential losses than to gains of equivalent magnitude (loss aversion), suggesting that the Brazilian market processes negative news about the studied companies with greater intensity than positive news of comparable content.

The results found are also compatible, in order of magnitude, with the negative abnormal returns reported by Dias and Malaquias (2021) for corruption events on the B3 (mean CAR of -2.29% in the 5-day window) — although the direct comparison is limited by the difference in the nature of the events (corruption news versus general news flow) and in the time window.

4.2.1. Results by company

Table 7 disaggregates the mean CAR by company and sentiment group, allowing an assessment of the heterogeneity of the estimated effect across the ten sample assets.

Table 7: Mean CAR by company and sentiment group

Ticker	Sentiment	N	Mean CAR (%)
ABEV3	NEGATIVE	167	-1.0366
ABEV3	NEUTRAL	468	-0.3433
ABEV3	POSITIVE	179	-0.3078
BBAS3	NEGATIVE	385	-0.1345
BBAS3	NEUTRAL	1,212	+0.1361
BBAS3	POSITIVE	407	-0.0215
BBDC4	NEGATIVE	625	-0.9769
BBDC4	NEUTRAL	1,048	+0.0109
BBDC4	POSITIVE	542	-0.1359
BPAC11	NEGATIVE	81	+0.0537
BPAC11	NEUTRAL	252	-0.0398
BPAC11	POSITIVE	273	+0.0519
ELET3	NEGATIVE	277	-0.2843
ELET3	NEUTRAL	336	+0.2212
ELET3	POSITIVE	246	+0.1109
ITSA4	NEGATIVE	27	+0.0528
ITSA4	NEUTRAL	43	-0.4177
ITSA4	POSITIVE	52	+0.1264
ITUB4	NEGATIVE	101	-0.0865
ITUB4	NEUTRAL	289	+0.0400
ITUB4	POSITIVE	177	+0.5299
PETR4	NEGATIVE	641	-0.7727
PETR4	NEUTRAL	959	-0.1253
PETR4	POSITIVE	603	+0.4187
VALE3	NEGATIVE	155	-0.2836
VALE3	NEUTRAL	212	-0.0675
VALE3	POSITIVE	163	+0.7194
WEGE3	NEGATIVE	22	-2.3470
WEGE3	NEUTRAL	70	-0.2629
WEGE3	POSITIVE	96	+1.3926

Source: Research data.

The disaggregation reveals relevant heterogeneity across assets. WEGE3 presents the largest-magnitude effects in both directions (+1.39% for positive news and -2.35% for negative news), although the reduced number of negative events for this asset (N=22) recommends caution in generalizing this specific result. BPAC11, on the other hand, presents a mean CAR close to zero in both directions, suggesting that the market prices news related to this institution quickly, or that the informational content of the news collected about it is, on average, less relevant.

A result that departs from the general pattern is that of ABEV3, whose POSITIVE news group is associated with a negative mean CAR (-0.31%). This pattern may be consistent with a “sell the news” effect, in which positive information had already been anticipated and priced in by the market before its formal disclosure, or it may reflect sampling noise given the moderate N (179) of this subgroup. This finding is revisited in the discussion of Section 4.3, in light of the firm fixed effects estimated in the logistic regression.

4.2.2. Limitations of the event-study analysis

Three methodological limitations must be acknowledged in interpreting the results of this section. First, the news sample presents relevant concentration in only three of the ten companies (Petrobras, Bradesco, and Banco do Brasil, which together represent 62% of the total), so the aggregate effect found in Table 6 may be partially influenced by the specific behavior of these three assets, even though the disaggregation by firm in Table 7 partially mitigates this concern by showing that the sign of the effect is qualitatively consistent across the majority of assets.

Second, the textual unit classified by the LLM was the news headline, and not the full body of the text, which privileges speed and processing cost at the expense of possible loss of nuance present in the complete content of the report.

Finally, when more than one news item about the same company is published on the same business day, all of them are associated with the same daily CAR value, even if they present different sentiments among themselves. This procedure, necessary given the daily granularity of the price data, tends to introduce noise into the relationship between sentiment and return, insofar as multiple potentially conflicting signals from the same day are compressed into a single return observation.

4.3. RESULTS: DIRECTIONAL ACCURACY AND LOGISTIC REGRESSION

Excluding the observations classified as NEUTRAL (per the criterion established in Section 3.6.1), the sample used in this section totals 5,219 news items. The majority-class baseline (proportion of $CAR > 0$ in the sample) corresponds to 51.64%, whereas the directional accuracy of Llama 3.1 8B was 54.74% — an improvement of 3.1 percentage points over the baseline.

McNemar's test confirms that this difference is not attributable to sampling variation ($\chi^2=9.4671$; $p=0.0021$), evidencing that the signal extracted by the model carries statistically detectable directional information, albeit of modest magnitude in absolute terms.

Table 8: Confusion matrix — model directional prediction

	Actual: CAR>0	Actual: CAR≤0
Predicted Positive	1,450	1,288
Predicted Negative	1,074	1,407

Source: Research data.

The confusion matrix does not reveal gross classification bias in either direction, with errors reasonably distributed between false positives and false negatives.

Table 9 presents the results of the logistic regression with firm fixed effects (Equation 8), taking ABEV3 as the reference category (omitted by requirement of model identification, given that categorical fixed effects require the exclusion of one level as the basis of comparison).

Table 9: Logistic regression — P(CAR>0) as a function of sentiment and firm fixed effects

Variable	Coefficient	Std. error	z	p-value
Intercept (ABEV3)	-0.4445	0.225	-1.975	0.048 **
BBAS3	+0.3680	0.252	1.461	0.144
BBDC4	+0.2268	0.259	0.875	0.381
BPAC11	+0.5494	0.307	1.789	0.074 *
ELET3	+0.3976	0.281	1.417	0.156
ITSA4	+0.4624	0.461	1.002	0.316
ITUB4	+0.4229	0.308	1.374	0.170
PETR4	+0.4372	0.266	1.646	0.100 *
VALE3	+0.6563	0.278	2.364	0.018 **
WEGE3	+0.4993	0.365	1.368	0.171
Sentiment (S)	+0.1844	0.040	4.608	<0.001 ***

Note: N=5,219. McFadden Pseudo R²=0.0107. Log-Likelihood=-3,575.9. Standard errors clustered by date. *p<0.10; **p<0.05; ***p<0.01. Reference category: ABEV3. Source: Research data.

The sentiment coefficient remains positive and statistically significant after clustering the standard errors by date ($z=4.608$; $p<0.001$), corresponding to an odds ratio of 1.2025: positive news multiplies by 1.20 the odds of a positive abnormal return, holding the firm fixed. The comparison with the standard errors without clustering ($SE=0.028$ vs. $SE=0.040$ with clustering, a 43% increase) evidences that cross-sectional dependence between simultaneous events does in fact exist and inflates the apparent precision of the estimators, but is not enough to compromise the significance of the parameter of interest.

The firm fixed effects largely lost their individual significance under the clustered specification, with the exception of VALE3 ($p=0.018$) and of PETR4 and BPAC11 at the 10% frontier. This pattern is expected: market shocks common within each event date introduce correlation among the errors of observations of distinct firms on the same day, inflating the standard errors of the firm coefficients more markedly than that of the sentiment coefficient, which varies more independently across dates. The loss of significance of the fixed effects does not alter the interpretation of the parameter of interest nor compromise the control function these effects perform in the model.

The Pseudo R^2 of 0.0107, although low in absolute terms, is compatible with the noisy nature of daily financial returns and with magnitudes reported in the literature for similar problems. ABEV3, by presenting the lowest base probability of a positive abnormal return among the ten companies (intercept = -0.4445), confirms the anomaly identified in Section 4.2.1, in which the positive-news group of this company was associated with a negative mean CAR.

Compared to the international literature, the directional accuracy found here (54.74%) is substantially lower than the hit rates reported by Lopez-Lira and Tang (2023) for GPT-4 (approximately 90% for the initial reaction) and than the 74.4% accuracy reported by Kirtac and Germano (2024) for the OPT model. This difference is compatible with the central finding of Lopez-Lira and Tang (2023) that financial-evaluation capacity is an emergent capability that increases with model scale: the Llama 3.1 8B used in this study is substantially smaller than GPT-4 and than the fine-tuned models employed by Kirtac and Germano (2024), which makes the result modest yet still statistically robust, coherent with the theoretical expectations of the literature.

4.4. RESULTS: LONG/SHORT PORTFOLIO

The long/short portfolio strategy described in Section 3.7 produced a position on 499 business days within the analyzed period. Table 10 compares the risk-adjusted performance of the strategy with the two benchmarks defined in the methodology.

Table 10: Sharpe ratio — long/short strategy versus benchmarks

Strategy	Sharpe	Std. error (Lo, 2002)	N days
Long/Short (LLM signal)	3.7961	0.7207	499
Ibovespa (buy-and-hold)	0.3724	—	667
Basket of 10 stocks (buy-and-hold)	0.8221	—	667

Source: Research data.

The strategy's mean daily return was 0.2665%, resulting in an annualized Sharpe ratio of 3.80 (approximate 95% CI: [2.38; 5.21]), substantially higher than the two benchmarks (Ibovespa: 0.37; equal-weighted basket: 0.82) and of the same order of magnitude as the Sharpe of 3.05 reported by Kirtac and Germano (2024) for the U.S. market with the OPT model.

This result, however, must be interpreted with considerable caution, for four reasons. First, the backtest does not incorporate transaction costs or stock-borrow costs for the short positions, both relevant in the context of the Brazilian market. Second, the cross-sectional diversification inherent in the portfolio construction (daily averages across multiple assets) tends to mechanically reduce the variance of portfolio returns, inflating the estimated Sharpe ratio — a known artifact in academic backtests of this nature. Third, the Lo (2002) standard error used presupposes independent and identically distributed returns; since there is temporal overlap between event windows of nearby firms and dates, the real uncertainty associated with the Sharpe estimate is likely greater than that reported in Table 10. Fourth, and most fundamentally, the simplification described in Section 3.7.1 — attributing the entirety of the two-day return to the signal date — does not correspond to a strict simulation of portfolio execution.

For these reasons, the result of Table 10 should be read as evidence that the signal extracted by the LLM carries economically relevant content in a frictionless backtest environment, and not as an estimate of the risk-adjusted return realizable under effective trading conditions in the Brazilian market.

5. CONCLUDING REMARKS

The present work investigated the effectiveness of an open-source Large Language Model (Llama 3.1 8B Instruct) in classifying the sentiment of Portuguese-language financial news and its impact on the short-term returns of ten of the leading stocks listed on the B3, between 2023 and 2024. The investigation was conducted across four complementary layers of evidence, of increasing complexity, whose results converge on the same general conclusion: the sentiment extracted by the LLM carries statistically detectable informational content about the subsequent returns of the analyzed stocks, albeit of economically modest magnitude.

In the first layer, the classic event study (Section 4.2) found cumulative abnormal returns statistically different from zero for the positive-news group (+0.19%; $p < 0.001$) and the

negative-news group (-0.61% ; $p < 0.001$), with the neutral group functioning adequately as a control ($p = 0.607$). The asymmetry in magnitude between the positive and negative effects — the negative effect is about 3.3 times larger — is consistent with the loss-aversion literature reviewed in Section 2.1.

In the second layer, the reformulation of the problem as binary directional classification (Section 4.3) found an accuracy of 54.74% against a fair baseline of 51.64%, a statistically significant difference by McNemar's test ($p = 0.0021$) and corroborated by a logistic regression with firm fixed effects, in which sentiment remained significant (odds ratio of 1.20; $p < 0.001$) even controlling for heterogeneity across assets. The modest magnitude of this result, relative to the substantially larger accuracies reported by Lopez-Lira and Tang (2023) and Kirtac and Germano (2024) for larger-scale models, is compatible with these authors' thesis that financial-reasoning capacity is an emergent capability that grows with model size.

In the third layer, the long/short portfolio backtest (Section 4.4) produced a Sharpe ratio of 3.80, higher than the market benchmarks considered, signaling potential economic relevance of the extracted signal — a result that, however, was qualified by substantial methodological caveats regarding the absence of transaction costs, the inflation of the Sharpe by cross-sectional diversification, and the possible underestimation of the standard error due to the presence of autocorrelation not captured by the Lo (2002) approximation.

Taken together, this evidence contributes to the scarce empirical literature on the use of LLMs applied to financial news in Portuguese, extending to the Brazilian context evidence documented mostly for the U.S. market, and dialogues with the national event-study literature applied to corporate news (DIAS; MALAQUIAS, 2021; PADULA; ALBUQUERQUE, 2018).

The limitations of this study — detailed in Sections 4.2.2, 3.6, and 3.7.1 — include the concentration of the news sample in three of the ten analyzed companies, the granularity of the classification at the headline level, the aggregation of multiple same-day news items into a single return value, and the simplifications inherent in the frictionless portfolio backtest. For future research, the following are suggested: (i) extending the classification to the full text of the reports; (ii) systematically comparing the performance of models of different scales in the Brazilian context, along the lines of Lopez-Lira and Tang (2023); (iii) incorporating

transaction costs and stock-borrow costs into the portfolio backtest; and (iv) expanding the sample of companies to reduce the sectoral imbalance identified in the news collection.

REFERENCES

- CAMERON, A. C.; MILLER, D. L. A Practitioner's Guide to Cluster-Robust Inference. *Journal of Human Resources*, v. 50, n. 2, p. 317–372, 2015.
- DE LONG, J. B.; SHLEIFER, A.; SUMMERS, L. H.; WALDMANN, R. J. Noise Trader Risk in Financial Markets. *Journal of Political Economy*, v. 98, n. 4, p. 703–738, 1990.
- DIAS, B.; MALAQUIAS, R. Corrupção e retorno de ações de empresas listadas na B3. *Brazilian Review of Finance (Online)*, Rio de Janeiro, v. 19, p. 123–150, 2021.
- FAMA, E. F. Efficient capital markets: a review of theory and empirical work. *The Journal of Finance*, v. 25, n. 2, p. 383–417, 1970.
- FAMA, E. F. Efficient Capital Markets: II. *The Journal of Finance*, v. 46, n. 5, p. 1575–1617, 1991.
- GUPTA, U. GPT-InvestAR: enhancing stock investment strategies through annual report analysis with Large Language Models. 2023. Available at: <<https://arxiv.org/abs/2309.03079>>. Accessed: June 12, 2026.
- KAHNEMAN, D.; TVERSKY, A. Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, v. 47, n. 2, p. 263–292, 1979.
- KIRTAC, K.; GERMANO, G. Enhanced financial sentiment analysis and trading strategy development using Large Language Models. In: *WORKSHOP ON COMPUTATIONAL APPROACHES TO SUBJECTIVITY, SENTIMENT, & SOCIAL MEDIA ANALYSIS*, 14., 2024. Proceedings [...]. p. 1–10.
- LO, A. W. The statistics of Sharpe ratios. *Financial Analysts Journal*, v. 58, n. 4, p. 36–52, 2002.
- LOPEZ-LIRA, A.; TANG, Y. Can ChatGPT forecast stock price movements? Return predictability and Large Language Models. *SSRN Electronic Journal*, 2023.
- LOUGHRAN, T.; MCDONALD, B. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, v. 66, n. 1, p. 35–65, 2011.
- MACKINLAY, A. C. Event studies in economics and finance. *Journal of Economic Literature*, v. 35, n. 1, p. 13–39, 1997.
- MCNEMAR, Q. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, v. 12, n. 2, p. 153–157, 1947.
- MILLER, D. An introductory guide to event study models. *Journal of Economic Perspectives*, v. 37, n. 2, p. 203–230, 2023.
- PADULA, A.; ALBUQUERQUE, P. H. M. Corrupção governamental no mercado de capitais: um estudo acerca da Operação Lava Jato. *RAE – Revista de Administração de Empresas, FGV EAESP*, v. 58, n. 4, p. 405–417, 2018.
- SHILLER, R. J. From Efficient Markets Theory to Behavioral Finance. *Journal of Economic Perspectives*, v. 17, n. 1, p. 83–104, 2003.

- SHLEIFER, A.; VISHNY, R. W. The Limits of Arbitrage. The Journal of Finance, v. 52, n. 1, p. 35–55, 1997.
- TETLOCK, P. C. Giving content to investor sentiment: the role of media in the stock market. The Journal of Finance, v. 62, n. 3, p. 1139–1168, 2007.
- VASWANI, A. et al. Attention is all you need. Advances in Neural Information Processing Systems, v. 30, 2017.

APPENDIX A – ROBUSTNESS TEST RESULTS

A.1 Cross-Sectional Dependence Test and OLS Regression with Date-Clustered Errors

To assess the existence of cross-sectional dependence among the cumulative abnormal returns (CAR) of simultaneous events, the pairwise covariance of the CARs of news published on the same day was computed. A total of 115,356 pairs of simultaneous events were identified, with a mean covariance of 0.000367 (t-stat = 50.76): statistically significant but economically negligible. As a complementary robustness analysis, an OLS regression of CAR on the sentiment dummies was estimated with standard errors clustered by date (as justified in Section 3.6.4). The results are presented in Table A.1.

Table A.1: OLS regression — CAR as a function of sentiment (date-clustered standard errors)

Variable	Coefficient	Std. error	z	p-value
Intercept (NEGATIVE)	−0.0061	0.002	−3.169	0.002 **
NEUTRAL	+0.0059	0.001	4.004	<0.001 ***
POSITIVE	+0.0079	0.002	4.351	<0.001 ***

Note: N=10,108. R²=0.014. Reference category: NEGATIVE. Standard errors clustered by date. **p<0.05; ***p<0.01.
Source: Research data.

The coefficients are statistically significant after clustering (NEGATIVE: z=−3.17, p=0.002; POSITIVE vs. NEGATIVE: z=4.35, p<0.001), confirming that the conclusions of Section 4.2 are robust to cross-sectional dependence. The implied mean CARs: NEGATIVE=−0.61%; NEUTRAL=−0.02%; POSITIVE=+0.18% — identical to the means of Table 6.

A.2 Logistic Regression without Clustering of Standard Errors (original specification)

For comparison with the specification adopted in Section 4.3, the logistic regression without clustering of standard errors is presented. The comparison allows quantifying the effect of cross-sectional dependence on the apparent precision of the estimators.

Table A.2: Logistic regression — P(CAR>0) without clustering of standard errors

Variable	Coefficient	Std. error	z	p-value
Intercept (ABEV3)	−0.4445	0.111	−4.022	<0.001 ***

Variable	Coefficient	Std. error	z	p-value
BBAS3	+0.3680	0.132	2.797	0.005 ***
BBDC4	+0.2268	0.125	1.809	0.070 *
BPAC11	+0.5494	0.155	3.556	<0.001 ***
ELET3	+0.3976	0.141	2.816	0.005 ***
ITSA4	+0.4624	0.252	1.837	0.066 *
ITUB4	+0.4229	0.164	2.585	0.010 **
PETR4	+0.4372	0.124	3.516	<0.001 ***
VALE3	+0.6563	0.158	4.147	<0.001 ***
WEGE3	+0.4993	0.216	2.308	0.021 **
Sentiment (S)	+0.1844	0.028	6.488	<0.001 ***

Note: N=5,219. McFadden Pseudo R²=0.0107. Log-Likelihood=-3,575.9. Non-clustered standard errors (standard MLE).

*p<0.10; **p<0.05; ***p<0.01. Reference category: ABEV3. Source: Research data.

The standard error of the sentiment coefficient increased from 0.028 (without clustering) to 0.040 with clustering by date (+43%), with the z-statistic reducing from 6.488 to 4.608. The firm fixed effects had their standard errors inflated by clustering, reflecting the intra-date correlation among observations of distinct firms. In both specifications, the sentiment coefficient remains highly significant (p<0.001), confirming the robustness of the main result of Section 4.3.